Chapter 1

Performance
- Algorithm – both source level statements and IO operations
- Programming Language, compiler and architecture – Determines no. of instructions for each source level statement
- Processor and memory system – how fast instructions may be executed
- I/O system (Hardware/OS)

7 great ideas in Computer Architecture

* (i) Use abstraction – dividing systems into layers to manage complexity

(ii) Make common case fast – optimizing frequently performed operations

(iii) Parallelism – running tasks simultaneously

(iv) Pipelining – a way to achieve parallelism

(v) Prediction – Anticipating future data/types

(vi) Hierarchy of memories – ~~op~~organizing memory into levels based on speed and size

(vii) Dependability via redundancy – Adding extra comp. to ensure reliability

1 → Application software – python

2 → system software
- Compiler: translates code into machine code
- OS: handling I/O, memory, storage, scheduling tasks.

3 → Hardware

levels of program code

High level language → print("Hello world")

↓ compiler ↓

Assembly language → slli x6, x11, 3

↓ Assembler ↓

Hardware representation → 0000 0001
1001 1100

Components of a computer
- Input
- Output
- memory
- Datapath } – processor
- Control }

Why silicon?
- The most available

Chapter 1

Response ~~a~~ Time → Execution time

& Throughput → Total work done per unit time

☆☆ The lesser the response time, the better the performance.

Relative performance

$$Perf_x = \frac{1}{ExecTime_x}$$

[$Comp_A$ completes a task in 10s.

$$P_A = \frac{1}{10}$$]

X is n times faster than y.

$$\frac{P_x}{P_y} = n$$

P_x → x better

↑ Perf ↑ Perf

↓ Exec ↑ Perf

$$\frac{\frac{1}{E_x}}{\frac{1}{E_y}} = n \Rightarrow \frac{E_y}{E_x} = n$$

↳ x better

CPU time — User CPU time / System CPU time

Clock cycle

Clock period/ clock duration/ clock cycle time → Time needed to complete one clock cycle

Clock frequency → Clock cycles completed per second.

$4\,GHz = 4 \times 10^9\,Hz$

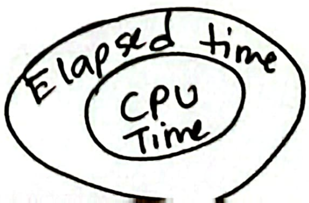
$$\boxed{\frac{1}{s} = Hz}$$

Elapsed time/ Response time/ Execution time } same

↳ total time to get work done

CPU Time

↳ CPU spends to compute a specific task



$$\boxed{1\,ps = 10^{-12}\,s}$$

CPU performance & its factors (Elapsed time = CPU time)

$$\text{CPU } \cancel{\text{exec}} \text{ time} = \frac{\text{CPU clock cycles}}{\text{clock rate}}$$

$$\text{CPU time} = \text{CPU clock cycles} \times \text{clock cycle time}$$

Chap 1 (formulas)

IC Cost Calculation

$$\text{Cost per die} = \frac{\text{Cost per wafer}}{\text{Dies per wafer} \times \text{Yield}}$$

$$\text{Dies per wafer} = \frac{\text{Wafer area}}{\text{Die area}}$$

Die Area = height X ~~w~~ width

wafer area = πr^2

Working dies per wafer = dies per wafer X yield

$$\text{Yield} = \frac{1}{(1 + (\text{defects per area} \times \text{die area}))^n}$$

n = given

$\frac{\text{die area}}{2}$, when n=2

$$\text{Performance} = \frac{1}{\text{Execution Time}}$$

CPU Time = ~~CP~~ Clock cycles X Clock ~~T~~ Cycle Time

$$CCT = \frac{1}{CR} \qquad = \frac{\text{Clock cycles}}{\text{Clock rate}}$$

$$CC = IC \times ACPI \quad = \frac{\text{Instruction count} \times \text{Avg CPI}}{\text{Clock rate}}$$

$$= IC \times \text{Avg CPI} \times \text{clock cycle time}$$

$$\text{Avg CPI} = \frac{\Sigma(\cancel{\text{Avg}}\ \text{CPI of each instruction type} \times \text{no. of instructions of that type})}{\text{Total no. of instructions}}$$

$$\text{SPEC Ratio} = \frac{\text{Reference Time}}{\text{CPU Time}}$$

$$\text{Geometric Mean} = \sqrt[n]{S.R_1 \times SR_2 \ldots \times SR_n}$$

$$T_{imp} = \frac{T_{aff}}{n} + T_{unaff} \qquad [n = \text{improv factor}]$$

$$\text{Power} = CL \times V^2 \times F = CV^2F$$

MIPS = 1 million instructions exectued in 1sec

~~perf IPS =~~

$$\text{Performance (IPS)} = \frac{\text{Clock rate}}{\text{CPI}}$$

$$\text{MIPS} = \frac{\text{Clock rate}}{\text{CPI} \times 10^6}$$

IC cost example

Q1

Given,

X
- 19 cm diameter wafer
- cost = 20
- contain = 89 dies
- defects = 0.023 defects/cm²

Y
- 20 cm diameter wafer
- cost = 15
- contain = 100 dies
- defects = 0.031 defects/cm²

n = 2

$$\text{Yield} = \frac{1}{(1+(\text{defects} \times \text{die area}))^2}$$

n = 2

die area = h × w

$$\text{dies per wafer} = \frac{\text{wafer area}}{\text{die area}}$$

$$\Rightarrow 89 = \frac{283.53}{\text{die area}}$$

$$\Rightarrow \text{die area} = \frac{283.53}{89}$$

$$= 3.19$$

wafer area $= \pi r^2$

$r = \frac{19}{2}$ cm

$= 9.5$ cm

W.A $= 9.5^2 \times \pi$

$= 283.53 \text{ cm}^2$

$$y = \frac{1}{\left(1+\left(0.023 \times \frac{3.19}{2}\right)\right)^2}$$

$\cancel{= 0.867} = \cancel{86.8\%} = 0.93 = 93\%$

for Y,

dies per wafer = $\dfrac{\text{wafer area}}{\text{die area}}$

$\Rightarrow dpw_y = \dfrac{WA_y}{DA_y}$

$WA_Y = \pi \times (10)^2$

$= 314.16 cm^2$

$dpw_y = 100$

$\Rightarrow 100 = \dfrac{314.16}{DA_y}$

$\Rightarrow DA_y = \dfrac{314.16}{100} = 31416$

$Y_y = \dfrac{1}{\left(1 + \left(\text{defect} \times \frac{dA_y}{2}\right)\right)^2}$

$= \dfrac{1}{\left(1 + \left(0.031 \times \frac{3.14}{2}\right)\right)^2}$

$= 0.909 = \cancel{90\%}\ 91\%$

≈ 0.91

b) Cost per die = $\dfrac{\text{Cost per wafer}}{\text{die } \cancel{\text{after}} \text{ per wafer} \times \text{yield}}$

$C_x = \dfrac{20}{89 \times 0.93} = 0.242$

$C_y = \dfrac{\cancel{20}\ 15}{100 \times 0.91} = 0.165$

Q2

5

Clock rate of PS = 2.7 GHz
Clock rate of Xbox = 3.0 GHz

	A	B	C	D
PS5	7	2	3	6
XboxX	5	4	2	1

for A,

$$\text{Avg CPI} = (7 \times 300 \times 10^3 + 2 \times 500 \times 10^3 + 3 \times 100 \times 10^3 + 6 \times 100 \times 10^3) \div 1 \times 10^6$$

$$= 4$$

$$\cancel{= (5 \times 10}$$

for B,

$$\text{Avg CPI} = (5 \times 3 \times 10^5 + 4 \times 5 \times 10^5 + 2 \times 1 \times 10^5 + 1 \times 1 \times 10^5) \div 1 \times 10^6$$

$$= 3.8$$

$$\frac{4}{3.8} = 1.05 \text{ or } 0.2 \text{ for PS5}$$

for PS5,

$$\text{CPU Time} = \frac{\text{Avg CPI} \times \text{IC}}{\text{CR}}$$

$IC = 1 \times 10^6$

$\text{Avg CPI} = 4$

$CR = 2.7 \times 10^9$

$$= \frac{4 \times 1 \times 10^6}{2.7 \times 10^9}$$

$$= 1.481 \text{ ms}$$

for Xbox X,

$$\text{CPU Time} = \frac{\text{Avg CPI} \times \text{IC}}{\text{CR}}$$

$CR = 3 \times 10^9$

$IC = 1 \times 10^6$

$CPI = 3.8$

$$= \frac{3.8 \times 1 \times 10^6}{3 \times 10^9}$$

$$= \frac{3.8}{3} \times 10^{-3}$$

$$= 1.267 \text{ ms}$$

$$\text{Diff} = 0.214 \text{ ms}$$

$$\text{SPEC Ratio} = \frac{\text{Ref time}}{\text{Exec time}} = \frac{120}{1.481} = 81.026$$

Q3

Sameinstruction set = I same

for P_1,

$$\text{CPU Time} = \frac{1.5I}{3\times10^9}$$

$$= 5\times10^{-10} I$$

for P_2,

$$\text{CPU Time} = \frac{\cancel{2.5}\,1\,I}{2.5\cancel{4}\times10^9}$$

$$= \cancel{6.25}\,4\times10^{-10}$$

for P_3,

$$\text{CPU Time} = \frac{2.2I}{4\times10^9}$$

$$= 5.5\times10^{-10}$$

most time = P_2

second most = P_3

~~P_1 and P~~

$\therefore P_1$ highest performance.

$P_1 > P_3 > P_2$

Q3

for P_1,

$$\text{CPU Time} = \frac{1.5I}{3\times10^9}$$

$$= 5\times10^{-10}$$

for P_2,

$$\text{CPU Time} = \frac{1I}{2.5\times10^9}$$

$$= 4\times10^{-10}$$

for P_3,

$$\text{CPU Time} = \frac{2.2I}{4\times10^9}$$

$$= 5.5\times10^{-10}$$

P_2 has best perf

$$\text{CPU Time} = \frac{IC \times CPI}{\text{Clock rate}}$$

$$\Rightarrow 10 = \frac{IC \times 1.5}{3 \times 10^9}$$

$$\Rightarrow IC = 2 \times 10^{10}$$

Again,

$$10 = CC \times CCT$$

$$\Rightarrow 10 = CC \times 3.33\dot{3} \times 10^{-10}$$

$$= 3 \times 10^{10} \text{ clock cycles}$$

$$CCT = \frac{1}{CR} = \frac{1}{3 \times 10^9} = 3.33 \times 10^{-10}$$

Similarly

for B,

$$10 = \frac{IC \times 1}{2.5 \times 10^9}$$

$$\Rightarrow IC = 2.5 \times 10^{10}$$

$$10 = CC \times CCT$$

$$10 = CC \times 4 \times 10^{-10}$$

$$CC = 2.5 \times 10^{10}$$

$$10 = \frac{IC \times 2.2}{4 \times 10^9}$$

$$\Rightarrow IC = 1.8\dot{1} \times 10^{10}$$

$$10 = CC \times 2.5 \times 10^{-10}$$

$$CC = 4 \times 10^{10}$$

4

Add	Sub	Sll	
70	85	40	= 195

7

$$\begin{array}{r} 250 \\ -195 \\ \hline 55\text{ s} \end{array}$$ others

New Time = $(70-14+85+40+55)$

$= 236$

$\therefore (250-236) = 14\text{s}$

5

P_1 — Clock rate = 4×10^9
Avg CPI = 0.9
IC = 5×10^9

P_2 — CR = 3×10^9
CPI = 0.75
IC = 1×10^9

$$\text{MIPS}_{P_1} = \frac{4 \times 10^9}{0.9 \times 10^6} = 4444.44$$

$$\text{MIPS}_{P_2} = \frac{3 \times 10^9}{0.75 \times 10^6} = 4000$$

$$\text{Cputime} = \frac{5 \times 10^9 \times 0.9}{4 \times 10^9} = 1.125\text{ s}$$

$$\text{CPU Time} = \frac{1 \times 10^9 \times 0.75}{3 \times 10^9} = 0.25\text{ s}$$

Time a = 10s

After 30% reduction = 7s

	Before CPI	After CPI	IC
P_1	1.5	1.8	2×10^{10}
P_2	1.0	1.2	2.5×10^{10}
P_3	2.2	2.64	1.81×10^{10}

$$\left.\begin{aligned} \text{CPU time} &= \frac{I \times 1.8}{\text{Rate}} \\ \Rightarrow 7 &= \frac{1.8 \times 2\cancel{3} \times 10^{10}}{\text{Rate}} \\ \Rightarrow CR &= \cancel{6.43 \times 10^9} = \cancel{6.43\,GHz} \\ &\quad 5.14 \times 10^9 = 5.14\,GHz \end{aligned}\right\} P_1$$

P_2,

$$7 = \frac{1.2 \times 2.5 \times 10^{10}}{\text{rate}}$$

$$\Rightarrow CR = 4.29 \times 10^9 = 4.29\,GHz$$

P_3,

$$CR = \frac{2.64 \times 1.81 \times 10^{10}}{7} = 6.9 \times 10^9$$

$$= 6.9\,GHz$$